Jessica Chen
HW6

Output for CountRecs.scala:

```
jhc10024_nyu_edu@nyu-dataproc-m:~$ spark-shell --deploy-mode client -i  CountRecs.scala
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/04 17:33:19 INFO SparkEnv: Registering MapOutputTracker
26/04/04 17:33:19 INFO SparkEnv: Registering BlockManagerMaster
26/04/04 17:33:19 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/04/04 17:33:19 INFO SparkEnv: Registering OutputCommitCoordinator
Spark context Web UI available at http://nyu-dataproc-m.c.hpc-dataproc-19b8.internal:39647
Spark context available as 'sc' (master = yarn, app id = application_1768446951072_8823).
Spark session available as 'spark'.
26/04/04 17:33:33 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
Total records: 7038863
Total rows (RDD count): 7038863
Distinct values for CAD_EVNT_ID:
+-----------+
|CAD_EVNT_ID|
+-----------+
|108645760  |
|108648602  |
|108653411  |
|108655229  |
|108655857  |
|108656825  |
|108659548  |
|108664095  |
|108664770  |
|108666708  |
|108667487  |
|108667820  |
|108668012  |
|108668650  |
|108669559  |
|108671717  |
|108672432  |
|108673521  |
|108676251  |
|108677244  |
+-----------+
only showing top 20 rows

Number of distinct values in CAD_EVNT_ID: 6858302

Distinct values for CREATE_DATE:
+-----------+
|CREATE_DATE|
+-----------+
|04/27/2025 |
|04/12/2025 |
|04/13/2025 |
|04/22/2025 |
|04/23/2025 |
|04/25/2025 |
|04/30/2025 |
|04/18/2025 |
|05/01/2025 |
|05/02/2025 |
|04/11/2025 |
|04/15/2025 |
|04/16/2025 |
|04/21/2025 |
|04/17/2025 |
```

```
|04/20/2025 |
|04/24/2025 |
|04/26/2025 |
|04/14/2025 |
|04/28/2025 |
+-----------+
only showing top 20 rows

Number of distinct values in CREATE_DATE: 365

Distinct values for INCIDENT_DATE:
+-------------+
|INCIDENT_DATE|
+-------------+
|04/27/2025   |
|04/12/2025   |
|04/13/2025   |
|04/22/2025   |
|04/23/2025   |
|04/25/2025   |
|04/30/2025   |
|04/18/2025   |
|05/01/2025   |
|05/02/2025   |
|04/11/2025   |
|04/15/2025   |
|04/16/2025   |
|04/21/2025   |
|04/17/2025   |
|04/20/2025   |
|04/24/2025   |
|04/26/2025   |
|04/14/2025   |
|04/28/2025   |
+-------------+
only showing top 20 rows

Number of distinct values in INCIDENT_DATE: 366

Distinct values for INCIDENT_TIME:
+-------------+
|INCIDENT_TIME|
+-------------+
|07:41:26     |
|11:33:36     |
|12:06:02     |
|12:08:47     |
|14:06:41     |
|15:38:36     |
|15:48:43     |
|19:41:08     |
|19:41:12     |
|20:28:47     |
|21:10:12     |
|22:52:37     |
|23:00:39     |
|23:06:22     |
|23:27:54     |
|01:19:22     |
|03:30:42     |
|05:45:39     |
|07:19:09     |
|08:31:10     |
+-------------+
only showing top 20 rows
```

```
Number of distinct values in INCIDENT_TIME: 86400

Distinct values for NYPD_PCT_CD:
+-----------+
|NYPD_PCT_CD|
+-----------+
|7          |
|18         |
|61         |
|75         |
|26         |
|77         |
|68         |
|60         |
|25         |
|115        |
|81         |
|33         |
|48         |
|106        |
|84         |
|9          |
|49         |
|63         |
|123        |
|94         |
+-----------+
only showing top 20 rows

Number of distinct values in NYPD_PCT_CD: 80

Distinct values for BORO_NM:
+-------------+
|BORO_NM      |
+-------------+
|BROOKLYN     |
|(null)       |
|QUEENS       |
|BRONX        |
|STATEN ISLAND|
|MANHATTAN    |
+-------------+

Number of distinct values in BORO_NM: 6

Distinct values for RADIO_CODE:
+----------+
|RADIO_CODE|
+----------+
|54C3      |
|39Q1      |
|54L1      |
|39V2      |
|69N2      |
|24W7      |
|10F1      |
|24Q7      |
|10V8      |
|12U1      |
|50N8      |
|24Q8      |
|12X       |
|33S5      |
|24S1      |
|10Q2      |
```

```
|59M       |
|54E1      |
|68M2      |
|32Q9      |
+----------+
only showing top 20 rows

Number of distinct values in RADIO_CODE: 496

Distinct values for TYP_DESC:
+-----------------------------------------------------+
|TYP_DESC                                             |
+-----------------------------------------------------+
|SEE COMPLAINANT: OTHER/OUTSIDE                       |
|INVESTIGATE/POSSIBLE CRIME: KNIFE/LTD ACC HWY        |
|DISPUTE: FIREARM/TRANSIT                             |
|OTHER CRIMES (IN PROGRESS): HARASSMENT/TRANSIT       |
|ASSAULT (PAST): SHOTS/INSIDE                         |
|OTHER NON-CRIME INCIDENT (SPECIFY): TRANSIT          |
|OTHER CRIMES (IN PROGRESS): HARASSMENT/SCHOOL        |
|INVESTIGATE/POSSIBLE CRIME: NARCO SALES/TRANSIT      |
|INVESTIGATE/POSSIBLE CRIME: SERIOUS OTHER/TRANSIT    |
|ROBBERY (IN PROGRESS): RESIDENCE                     |
|PO/SECURITY HOLDING SUSPECT: CIV CLOTHES/INSIDE      |
|LARCENY (IN PROGRESS): VEHICLE/LTD ACC HWY           |
|EMS REQ ASSIST/OUTSIDE                               |
|ASSAULT (IN PROGRESS): WEAPON/CHILD                  |
|AMBULANCE CASE: SEVERED LIMB/TRANSIT                 |
|ALL OUT DIRECTED                                     |
|ASSIST POLICE OFFICER: CIV CLOTHES/OUTSIDE           |
|SEE COMPLAINANT: 85 - MOS/INSIDE                     |
|OTHER NON-CRIME INCIDENT (SPECIFY)                   |
|OTHER CRIMES (IN PROGRESS): VIOL ORDER PROTECT/TRANSIT|
+-----------------------------------------------------+
only showing top 20 rows

Number of distinct values in TYP_DESC: 493

Distinct values for CIP_JOBS:
+------------+
|CIP_JOBS    |
+------------+
|Non CIP     |
|Critical    |
|Non Critical|
|Serious     |
+------------+

Number of distinct values in CIP_JOBS: 4

Distinct values for ADD_TS:
+---------------------+
|ADD_TS               |
+---------------------+
|01/01/2025 01:39:06 AM|
|01/01/2025 02:22:52 AM|
|01/01/2025 02:58:00 AM|
|01/01/2025 03:47:57 AM|
|01/01/2025 06:55:57 AM|
|01/01/2025 07:42:18 AM|
|01/01/2025 12:30:47 PM|
|01/01/2025 02:19:51 PM|
|01/01/2025 04:08:01 PM|
|01/01/2025 05:59:53 PM|
|01/01/2025 07:23:39 PM|
```

```
|01/01/2025 11:40:40 PM|
|01/02/2025 12:14:05 AM|
|01/02/2025 12:34:58 AM|
|01/02/2025 04:14:01 AM|
|01/02/2025 08:58:09 AM|
|01/02/2025 10:18:57 AM|
|01/02/2025 10:32:15 AM|
|01/02/2025 10:47:41 AM|
|01/02/2025 11:04:30 AM|
+----------------------+
only showing top 20 rows

Number of distinct values in ADD_TS: 6155563

Distinct values for DISP_TS:
+----------------------+
|DISP_TS               |
+----------------------+
|01/01/2025 02:18:55 AM|
|01/01/2025 01:39:06 AM|
|01/01/2025 02:58:00 AM|
|01/01/2025 03:13:46 AM|
|01/01/2025 03:58:25 AM|
|01/01/2025 06:17:06 AM|
|01/01/2025 06:55:57 AM|
|01/01/2025 07:42:18 AM|
|01/01/2025 09:45:55 AM|
|01/01/2025 03:54:01 PM|
|01/01/2025 04:13:33 PM|
|01/01/2025 05:50:31 PM|
|01/01/2025 07:23:39 PM|
|01/01/2025 09:08:26 PM|
|01/01/2025 09:28:47 PM|
|01/02/2025 12:14:05 AM|
|01/02/2025 12:28:32 AM|
|01/02/2025 07:01:43 AM|
|01/02/2025 07:15:55 AM|
|01/02/2025 08:58:09 AM|
+----------------------+
only showing top 20 rows

Number of distinct values in DISP_TS: 6261615

Distinct values for ARRIVD_TS:
+----------------------+
|ARRIVD_TS             |
+----------------------+
|01/01/2025 01:39:06 AM|
|01/01/2025 02:58:00 AM|
|01/01/2025 06:46:52 AM|
|01/01/2025 06:55:57 AM|
|01/01/2025 07:42:18 AM|
|01/01/2025 08:46:22 AM|
|01/01/2025 04:08:01 PM|
|01/01/2025 06:08:54 PM|
|01/01/2025 07:23:39 PM|
|01/01/2025 09:47:18 PM|
|01/02/2025 07:28:11 AM|
|01/02/2025 08:08:48 AM|
|01/02/2025 08:58:09 AM|
|01/02/2025 02:01:40 PM|
|01/02/2025 02:15:27 PM|
|01/02/2025 10:13:38 PM|
|01/02/2025 11:56:55 PM|
|01/03/2025 12:13:32 AM|
```

```
|01/03/2025 04:10:54 AM|
|01/03/2025 04:26:05 AM|
+----------------------+
only showing top 20 rows

Number of distinct values in ARRIVD_TS: 5009461

Distinct values for CLOSNG_TS:
+----------------------+
|CLOSNG_TS             |
+----------------------+
|01/01/2025 02:18:55 AM|
|01/01/2025 01:39:06 AM|
|01/01/2025 03:13:46 AM|
|01/01/2025 08:42:56 AM|
|01/01/2025 07:34:15 AM|
|01/01/2025 12:55:59 PM|
|01/01/2025 02:47:36 PM|
|01/01/2025 06:50:10 PM|
|01/01/2025 05:37:53 PM|
|01/01/2025 06:09:57 PM|
|01/01/2025 06:06:15 PM|
|01/01/2025 07:33:29 PM|
|01/01/2025 09:01:58 PM|
|01/01/2025 09:41:58 PM|
|01/01/2025 10:31:00 PM|
|01/02/2025 01:01:08 AM|
|01/02/2025 08:08:48 AM|
|01/02/2025 08:38:05 AM|
|01/02/2025 08:58:09 AM|
|01/02/2025 10:13:57 AM|
+----------------------+
only showing top 20 rows

Number of distinct values in CLOSNG_TS: 6253447

Total distinct RADIO_CODE-TYP_DESC pairs: 496
TYP_DESC values mapping to multiple RADIO_CODEs:
+---------------------------------------+------------+---------+
|TYP_DESC                               |RADIO_CODES |num_codes|
+---------------------------------------+------------+---------+
|FIRE: SCHOOL                           |[59I5, 59I] |2        |
|INVESTIGATE/POSSIBLE CRIME: OTHER/SCHOOL|[10Q5, 10Y5]|2        |
|VISIBILITY PATROL: WARRANT CHECK       |[75W, 75W9] |2        |
+---------------------------------------+------------+---------+

RADIO_CODEs with missing TYP_DESC:
+----------+--------+----------------------------------------->   (14 +
|RADIO_CODE|TYP_DESC|
+----------+--------+
+----------+--------+

RADIO_CODE values mapping to multiple TYP_DESC:
+----------+---------+---------+
|RADIO_CODE|TYP_DESCS|num_descs|
+----------+---------+---------+
+----------+---------+---------+

TYP_DESC values mapping to multiple CIP_JOBS:
+--------+------------+------------+
|TYP_DESC|CIP_JOBS_set|num_CIP_JOBS|
+--------+------------+------------+
+--------+------------+------------+
```

```
All TYP_DESC → CIP_JOBS mappings:
+-----------------------------+------------+
|TYP_DESC                     |CIP_JOBS_set|
+-----------------------------+------------+
|10-53 NO RMP REQUIRED        |[Non CIP]   |
|311 REFERRAL                 |[Non CIP]   |
|ALARMS: AUDIBLE/INSIDE       |[Non CIP]   |
|ALARMS: AUDIBLE/LTD ACC HWY  |[Non CIP]   |
|ALARMS: AUDIBLE/OUTSIDE      |[Non CIP]   |
|ALARMS: AUDIBLE/SCHOOL       |[Non CIP]   |
|ALARMS: AUDIBLE/TRANSIT      |[Non CIP]   |
|ALARMS: BANK/BURGLARY        |[Non CIP]   |
|ALARMS: BANK/HOLDUP          |[Non CIP]   |
|ALARMS: BANK/TRANSIT         |[Non CIP]   |
|ALARMS: COMMERCIAL/BURGLARY  |[Non CIP]   |
|ALARMS: COMMERCIAL/HOLDUP    |[Non CIP]   |
|ALARMS: COMMERCIAL/LTD ACC HWY|[Non CIP]  |
|ALARMS: COMMERCIAL/TRANSIT   |[Non CIP]   |
|ALARMS: OTHER/BURGLARY       |[Non CIP]   |
|ALARMS: OTHER/HOLDUP         |[Non CIP]   |
|ALARMS: OTHER/TRANSIT        |[Non CIP]   |
|ALARMS: RESIDENTIAL/ HOLDUP  |[Non CIP]   |
|ALARMS: RESIDENTIAL/BURGLARY |[Non CIP]   |
|ALL OUT DIRECTED             |[Non CIP]   |
+-----------------------------+------------+
only showing top 20 rows

Duplicate CAD_EVNT_ID rows:
+-----------+-----+
|CAD_EVNT_ID|count|
+-----------+-----+
|110973063  |2    |
|111030564  |2    |
|110980017  |2    |
|111106968  |2    |
|111137012  |2    |
|111149548  |2    |
|111158223  |2    |
|111277166  |2    |
|111308336  |2    |
|111376229  |2    |
|111435630  |2    |
|111462660  |2    |
|111505846  |2    |
|110991282  |2    |
|110997170  |2    |
|111055178  |2    |
|111059448  |2    |
|111188405  |2    |
|111210510  |2    |
|111251489  |2    |
+-----------+-----+
only showing top 20 rows

Min/Max CREATE_DATE and INCIDENT_DATE:
+---------------+---------------+-----------------+-----------------+
|min_CREATE_DATE|max_CREATE_DATE|min_INCIDENT_DATE|max_INCIDENT_DATE|
+---------------+---------------+-----------------+-----------------+
|2025-01-01     |2025-12-31     |2024-12-31       |2025-12-31       |
+---------------+---------------+-----------------+-----------------+

Max length of TYP_DESC:
+----------------+
|max_TYP_DESC_len|
+----------------+
|58              |
+----------------+
```

```
First 10 rows where INCIDENT_TS is more than 1 minute after ADD_TS:
+-----------+-----------+-----------+-----------+-----------+---------+-----------+---------+---------+---
```

Hive table creation & DESCRIBE output:

```
jhc10024_nyu_edu@nyu-dataproc-m:~$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/tez/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/tez/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
Hive Session ID = 078e45fc-f021-4e2e-bd11-21bf4dd73be3

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.apache.hadoop.hive.common.StringInternUtils (file:/usr/lib/hive/lib/hive-common-3.1.3.jar) to field java.net.URI.string
WARNING: Please consider reporting this to the maintainers of org.apache.hadoop.hive.common.StringInternUtils
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Hive Session ID = 8748b8d3-541b-4b03-bfd6-119c34ccaa29
hive> SET hive.metastore.warehouse.dir=/user/jhc10024_nyu_edu/hive/warehouse;
hive> CREATE DATABASE IF NOT EXISTS nypd;
OK
Time taken: 5.411 seconds
hive> SET hive.metastore.warehouse.dir=/user/jhc10024_nyu_edu/hive/warehouse;
hive> USE nypd;
OK
Time taken: 0.41 seconds
hive> CREATE EXTERNAL TABLE IF NOT EXISTS nypd_cfs_cleaned (
    >   CAD_EVNT_ID STRING,
    >   CREATE_DATE DATE,
    >   INCIDENT_DATE DATE,
    >   INCIDENT_TIME STRING,
    >   NYPD_PCT_CD INT,
    >   BORO_NM STRING,
    >   RADIO_CODE STRING,
    >   TYP_DESC STRING,
    >   TYP_DESC_ENUM BIGINT,
    >   CIP_JOBS STRING,
    >   ADD_TS TIMESTAMP,
    >   DISP_TS TIMESTAMP,
    >   ARRIVD_TS TIMESTAMP,
    >   CLOSNG_TS TIMESTAMP,
    >   RESPONSE_TIME STRING,
    >   DISPATCH_DELAY STRING,
    >   TRAVEL_TIME STRING
    > )
    > STORED AS PARQUET
    > LOCATION 'hdfs:///user/jhc10024_nyu_edu/hw6/nypd_cfs_cleaned';
OK
Time taken: 1.483 seconds
```

```
hive> SELECT * FROM nypd_cfs_cleaned LIMIT 10;
OK
112093677       2025-05-27      2025-05-27      13:35:21        46      BRONX   10H1    INVESTIGATE/POSSIBLE CRIME: CALLS FOR HELP/INSIDE       253     Non CIP 2025-05-27 13:35:21     2025-05-27 15:47:49     NULL    2025-05-27 22:47:47     NULL    02:12:28        NULL
110641418       2025-03-28      2025-03-28      08:52:40        78      BROOKLYN        53S     VEHICLE ACCIDENT: SPECIAL CONDITION     480     Non CIP 2025-03-28 08:52:40     2025-03-28 08:57:41     2025-03-28 09:45:31     2025-03-28 09:45:35     00:52:51        00:05:01        00:47:50
110637414       2025-03-28      2025-03-28      02:02:51        106     QUEENS  75M     TRAIN RUN/MOBILE ORDER MAINTENANCE SWEEP        463     Non CIP 2025-03-28 02:02:51     2025-03-28 02:02:51     2025-03-28 02:02:51     2025-03-28 04:04:18     00:00:00        00:00:00        00:00:00
116179035       2025-11-14      2025-11-14      19:19:11        106     QUEENS  53D     VEHICLE ACCIDENT: DISPUTE       472     Non CIP 2025-11-14 19:19:11     2025-11-14 19:33:37     2025-11-14 20:52:44     2025-11-14 21:00:51     01:33:33        00:12:26        01:21:07
113461461       2025-07-29      2025-07-29      11:36:30        72      BROOKLYN        10Y3    INVESTIGATE/POSSIBLE CRIME: SERIOUS/OTHER       286     Non CIP 2025-07-29 11:36:30     2025-07-29 11:51:39     NULL    2025-07-29 12:24:57     NULL    00:15:09        NULL
114200843       2025-08-23      2025-08-23      14:53:04        111     QUEENS  10Y8    INVESTIGATE/POSSIBLE CRIME: SERIOUS OTHER/LTD ACC HWY  284     Non CIP 2025-08-23 14:53:04     2025-08-23 15:49:34     2025-08-23 16:18:33     2025-08-23 16:29:39     01:25:29        00:56:30
00:28:59
113763632       2025-08-04      2025-08-04      16:17:04        45      BRONX   42      QUALITY OF LIFE COMPLAINT       402     Non CIP 2025-08-04 16:17:04     2025-08-04 16:17:04     2025-08-04 16:17:04     2025-08-04 16:28:46     00:00:00        00:00:00        00:00:00
113537701       2025-07-25      2025-07-25      20:41:36        78      BROOKLYN        75S     STATION INSPECTION BY TRANSIT BUREAU PERSONNEL 444      Non CIP 2025-07-25 20:41:36     2025-07-25 20:41:36     2025-07-25 20:41:36     2025-07-25 20:49:49     00:00:00        00:00:00
00:00:00
116094784       2025-11-11      2025-11-11      03:03:13        60      BROOKLYN        75O     TRAIN ORDER MAINTENANCE SWEEP (TOMS)    462     Non CIP 2025-11-11 03:03:13     2025-11-11 03:03:13     2025-11-11 03:03:13     2025-11-11 03:30:37     00:00:00        00:00:00        00:00:00
110195551       2025-03-09      2025-03-09      10:18:45        81      BROOKLYN        75D     VISIBILITY PATROL: DIRECTED     485     Non CIP 2025-03-09 10:18:45     2025-03-09 10:18:45     2025-03-09 10:18:45     2025-03-09 11:17:10     00:00:00 00:00:00       00:00:00
Time taken: 0.68 seconds, Fetched: 10 row(s)
```

```
hive> DESCRIBE nypd_cfs_cleaned;
OK
cad_evnt_id             string
create_date             date
incident_date           date
incident_time           string
nypd_pct_cd             int
boro_nm                 string
radio_code              string
typ_desc                string
typ_desc_enum           bigint
cip_jobs                string
add_ts                  timestamp
disp_ts                 timestamp
arrivd_ts               timestamp
closng_ts               timestamp
response_time           string
dispatch_delay          string
travel_time             string
Time taken: 0.526 seconds, Fetched: 17 row(s)
```

To access the Hive Table

Data location:
hdfs:///user/jhc10024_nyu_edu/hw6/nypd_cfs_cleaned

Run the following commands while in HIVE CLI:
hive> SET hive.metastore.warehouse.dir=/user/jhc10024_nyu_edu/hive/warehouse;
hive> USE nypd;
hive> SELECT * FROM nypd_cfs_cleaned LIMIT 10;